

VALDOSTA, GA, USA

WMO: 722166

Lat: 30.783N

Lon: 83.277W

Elev: 198

StdP: 14.59

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93845

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.6	30.6	12.6	10.5	37.8	17.9	13.6	40.7	18.6	58.6	16.8	55.5	2.2	0	0.347

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	96.5	76.6	94.4	76.3	92.6	75.9	80.3	89.8	79.3	88.7	78.4	87.5	5.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.0	146.8	83.1	77.0	141.8	82.5	76.1	137.1	81.8	43.7	89.7	42.8	88.4	41.9	88.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.9	13.0	11.7	21.8	99.1	3.2	2.7	19.5	101.0	17.7	102.6	15.9	104.1	13.6	106.1		
			19.9	82.2	3.4	1.6	17.4	83.4	15.5	84.3	13.5	85.2	11.1	86.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.3	51.9	56.0	61.3	67.3	74.9	80.3	82.2	82.0	78.2	69.7	60.0	55.1
	DBStd	12.67	9.49	8.94	8.13	6.36	4.94	3.39	2.63	2.73	3.93	6.95	8.04	9.47
	HDD50	216	94	39	13	1	0	0	0	0	0	0	12	57
	HDD65	1456	413	271	165	49	3	0	0	0	0	38	189	328
	CDD50	6895	154	205	363	520	771	910	998	992	845	610	313	214
	CDD65	2663	8	18	50	118	309	460	533	527	396	183	41	20
	CDH74	25024	47	131	477	1230	3126	4429	5346	5090	3346	1416	285	101
	CDH80	10783	2	11	82	356	1349	2145	2604	2378	1411	404	35	6
(14) Wind	WSAvg	4.6	5.0	5.0	5.2	5.0	4.6	4.1	3.7	3.9	4.7	4.4	4.6	4.6
(15) Precipitation	PrecAvg	48.3	4.1	4.0	4.3	4.0	3.1	5.6	5.6	5.1	4.1	2.8	2.2	3.2
	PrecMax	72.4	20.3	8.4	11.5	9.6	6.5	12.5	8.9	8.5	14.0	11.7	6.0	5.6
	PrecMin	35.7	0.8	0.7	0.1	0.6	0.5	1.3	2.8	1.3	1.2	0.0	0.0	1.4
	PrecStd	9.4	3.7	1.7	2.9	2.2	1.5	3.1	1.8	1.7	3.3	2.6	1.6	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.9	81.7	85.6	90.0	95.6	99.4	98.9	99.1	95.2	89.8	83.8	80.8
		MCWB	66.8	66.6	66.6	70.1	73.3	76.2	78.1	78.1	74.5	73.6	70.9	68.7
	2%	DB	74.9	78.5	82.2	86.7	92.2	95.9	95.7	95.3	92.3	86.9	80.7	76.9
		MCWB	64.7	65.6	65.9	68.3	72.1	75.9	77.2	77.8	74.6	72.0	68.8	66.8
	5%	DB	71.7	74.7	79.4	84.2	90.1	93.3	93.7	93.0	90.2	84.4	77.4	73.4
		MCWB	63.0	63.7	64.3	67.2	71.1	75.2	77.1	77.4	74.5	71.1	66.0	66.1
	10%	DB	67.7	71.2	76.3	81.5	87.7	90.7	91.5	90.9	88.1	81.9	74.1	70.3
		MCWB	61.5	61.5	63.3	66.2	70.6	75.2	76.9	77.1	74.3	69.9	65.4	64.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.4	70.2	71.7	73.6	77.1	80.4	81.9	81.4	79.3	77.3	74.9	72.5
		MCDB	74.0	76.5	78.3	82.1	87.6	90.6	91.3	91.1	85.6	83.7	79.4	75.5
	2%	WB	67.2	68.2	69.6	72.0	75.5	79.0	80.4	80.2	78.1	75.7	72.4	70.1
		MCDB	72.1	74.1	76.5	80.6	84.4	89.0	90.0	89.7	85.5	82.2	76.5	73.7
	5%	WB	65.1	66.4	67.9	70.5	74.3	78.0	79.3	79.2	77.2	74.4	70.0	67.9
		MCDB	69.6	72.1	74.9	78.9	83.5	87.8	88.9	88.6	85.0	80.3	74.3	72.2
	10%	WB	62.8	64.2	66.0	69.0	73.1	77.0	78.3	78.3	76.3	72.9	67.5	65.5
		MCDB	66.5	69.1	73.1	77.1	82.4	86.4	87.6	87.1	83.6	78.4	72.7	69.9
(35) Mean Daily Temperature Range	5% DB	MDBR	22.6	23.0	24.3	24.8	23.8	20.4	19.7	18.8	19.0	21.9	23.6	21.8
		MCDBR	24.7	24.9	26.8	27.2	26.3	24.3	22.2	21.5	22.2	22.5	24.4	22.6
		MCWBR	15.4	13.9	12.7	11.2	9.2	7.9	7.3	7.0	7.9	10.1	13.0	13.9
	5% WB	MCDBR	20.6	20.1	21.1	21.5	21.4	20.0	19.8	19.2	17.5	17.1	18.0	18.6
		MCWBR	14.7	12.9	11.9	11.1	8.6	7.8	7.5	7.1	7.0	8.5	11.6	13.0
(40) Clear-Sky Solar Irradiance	taub		0.327	0.335	0.359	0.387	0.425	0.471	0.508	0.492	0.432	0.378	0.348	0.332
	taud		2.519	2.495	2.449	2.357	2.291	2.226	2.126	2.172	2.330	2.455	2.483	2.532
	Ebn at Noon		287	295	293	287	275	261	251	254	267	276	277	279
	Edh at Noon		28	31	35	39	43	45	50	47	39	32	29	26
(44) All-Sky Solar Radiation	RadAvg		986	1174	1546	1875	2049	1894	1898	1718	1554	1371	1099	867
	RadStd		67	139	129	142	126	128	115	115	127	121	77	77

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115
(47) Regional (30 neighbors)		+0.92	+1.44	N/A	N/A	N/A	N/A	-40	-146	+306	+212

Nomenclature: See separate page